

BLE VITALS SCANNER

Orbytring Brand Edition

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Applied Position: Flutter Intern

Target Assignment: BLE Vitals Scanner (sensio / orbytring)

The Challenge & Objective

TECHNICAL MISSION STATEMENT

- **Objective:** Build a responsive, crash-free Flutter application that dynamically scans for nearby BLE signals, connects to a selected target device, and visualizes live continuous telemetry.
- **Core Constraint:** Rely exclusively on `flutter_reactive_ble` for native Bluetooth channels, permissions verification, and characteristic notification streams.
- **Design Strategy:** Reject default UI templates in favor of a brand-centric **White Theme** modeled after orbytring.com. Utilizes champagne gold and rose gold accents over metallic slate surfaces.
- **Reliability:** Verify and package a tree-shaken release build (APK) that parses raw vital streams correctly and runs warning-free.

Seamless Clinical User Journey

FIVE-STAGE HIGH-FIDELITY UX FLOW

- 1. Permissions Gate** — Checks and requests BLE & Location permissions at runtime, dynamically adapting states for varying Android SDK targets.
- 2. Visual Scanning** — Triggers high-performance radar sweeps to scan for nearby transceivers, ordering results descending by decibel RSSI strength.
- 3. Handshake Link** — Allows connecting to target BLE items, displaying status indicators and load sequences to represent network handshake states.
- 4. Service Tree Explorer** — Explores the full GATT services and characteristics database, translating complex hex UUIDs into clinical text descriptors (e.g. Heart Rate Service).
- 5. Live Vitals Stream** — Visualizes real-time subscriptions with a canvas-based Bezier ECG line graph, pulsing heart, and live hexadecimal byte logs.

Curated Brand Design System

PREMIUM ORBYTRING BRAND TOKENS

- **Canvas Background:** Hex **#F8FAFC** (Metallic Soft White background providing clinical cleanliness).
- **Surface Elements:** Hex **#FFFFFF** (Pure White card layouts lined with metallic silver borders).
- **Auric Accent Gold:** Hex **#C5A059** (Signature Orbyt Gold branding for radar sweeps, toggles, buttons, and status icons).
- **Vitals Rose Gold:** Hex **#D49A8F** (Orbyt Rose Gold for scrolling ECG lines, nodes, and pulsing heart visualizations).
- **Slate Typography:** Hex **#0F172A** (Stellar Black primary headers) & **#64748B** (Gunmetal Grey secondary parameter data).

Robust Architecture Blueprint

DECOUPLED SERVICE REPOSITORY DESIGN

- **BleService Interface:** Creates an abstract contract defining all required BLE properties, removing direct hardware vendor ties.
- **ReactiveBleService:** Concrete implementation powering the native radio transceiver using the *flutter_reactive_ble* stream API.
- **MockBleService:** Complete offline simulation replacement generating RSSI fluctuations, connection states, and realistic Heart Rate data.
- **Flexible Providers:** State architecture leverages *BleProvider* to manage scanning and simulator switches, while *VitalsProvider* listens to live characteristic notifications.

Dual Simulation Ecosystem

FLAWLESS VALIDATION ON EMULATORS & REAL HARDWARE

- **In-App Software Mode:** Enabled by a custom card switch at the top of the scan menu. Allows direct validation of ECG charts, GATT trees, and raw logs inside virtual android emulators.
- **External Python Peripheral Simulator (ble_simulator.py):** Built using *bless* and *bleak* libraries. Simulates a physical vital sensor broadcasting a standard Heart Rate service (0x180D) and characteristic (0x2A37).
- **Wander Generator:** Simulates natural heart rate drift (BPM drift with minor random noise) to verify clinical ECG chart scaling.

GATT Specification Vitals Parsing

COMPLIANT BINARY TELEMETRY EXTRACTION

Extracts live Beats Per Minute directly from standard raw binary packets (GATT 2A37):

```
// Raw heart rate value is parsed by inspecting flag bits:
if (packet.isNotEmpty) {
    final int flags = packet[0];
    final int bpm;

    if ((flags & 0x01) == 0) {
        // 8-bit Heart Rate value format
        bpm = packet[1];
    } else {
        // 16-bit Heart Rate value format (little-endian representation)
        bpm = packet[1] | (packet[2] << 8);
    }
    _updateVitals(bpm);
}
```

Rigorous Code Verification

COMPILER INTEGRITY & BUILD VALIDATION

- **Static Analyzer:** Executed *flutter analyze* reporting **No issues found!** with zero errors, zero warnings, and zero lints.
- **Widget Smoke Test:** Includes dynamic layout checks that test initially mapped screens.
- **Production Release Build:** Compiled successfully using *flutter build apk --release*.
- **Delivery Output:** Highly optimized, tree-shaken Android APK compiled at **build\app\outputs\flutter-apk\app-release.apk** (44.2MB).

THANK YOU

BLE Vitals Scanner Submission

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